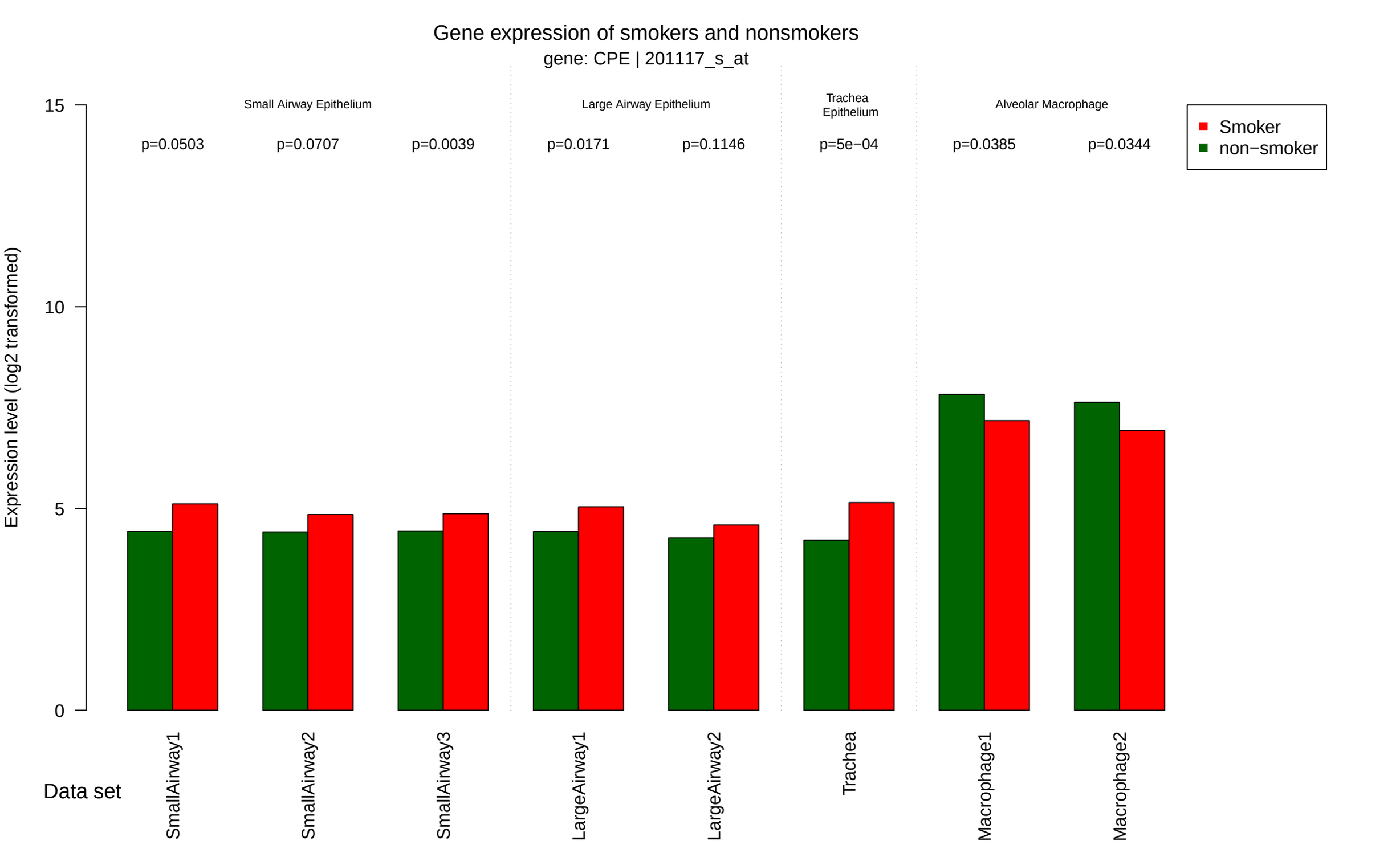
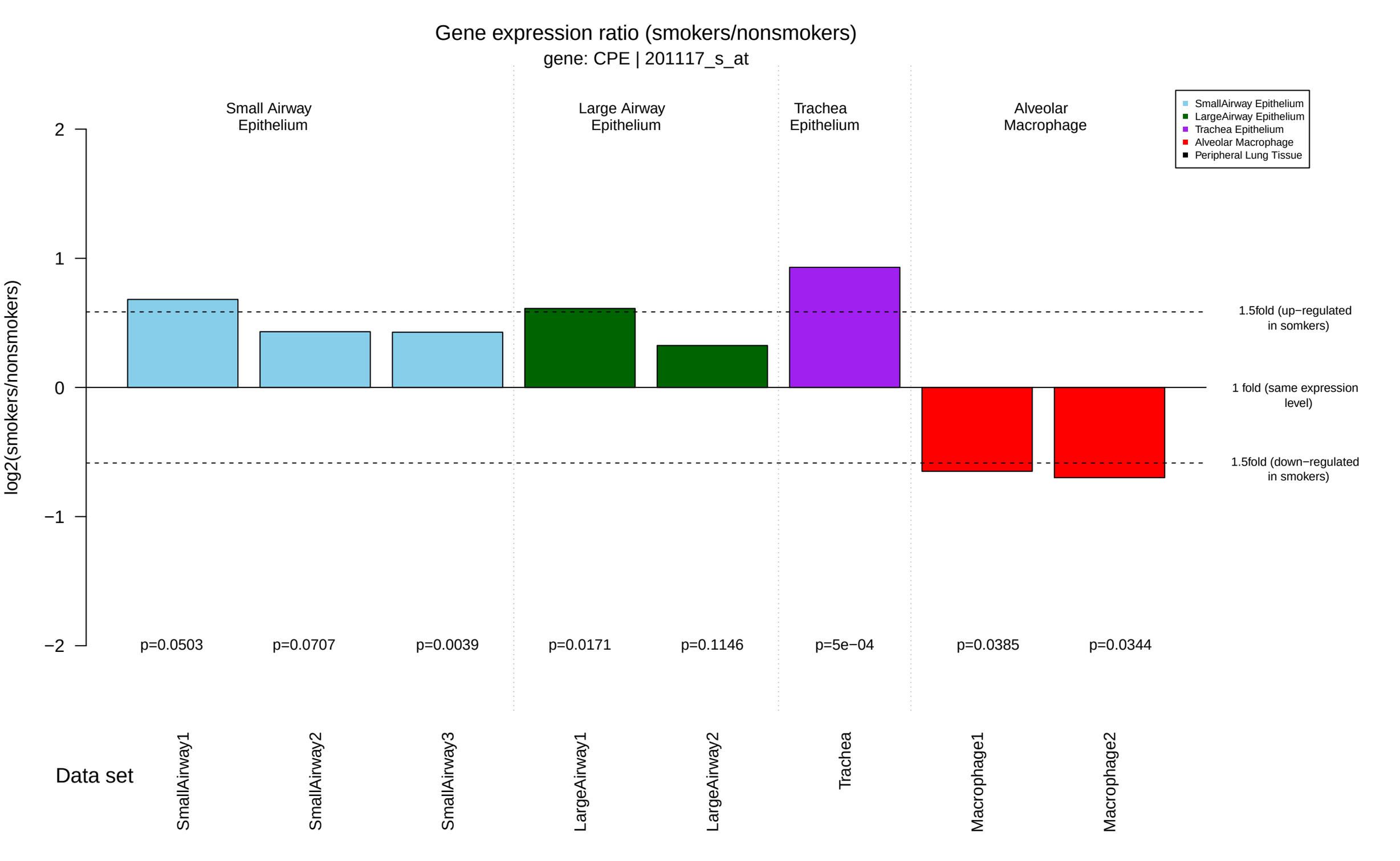
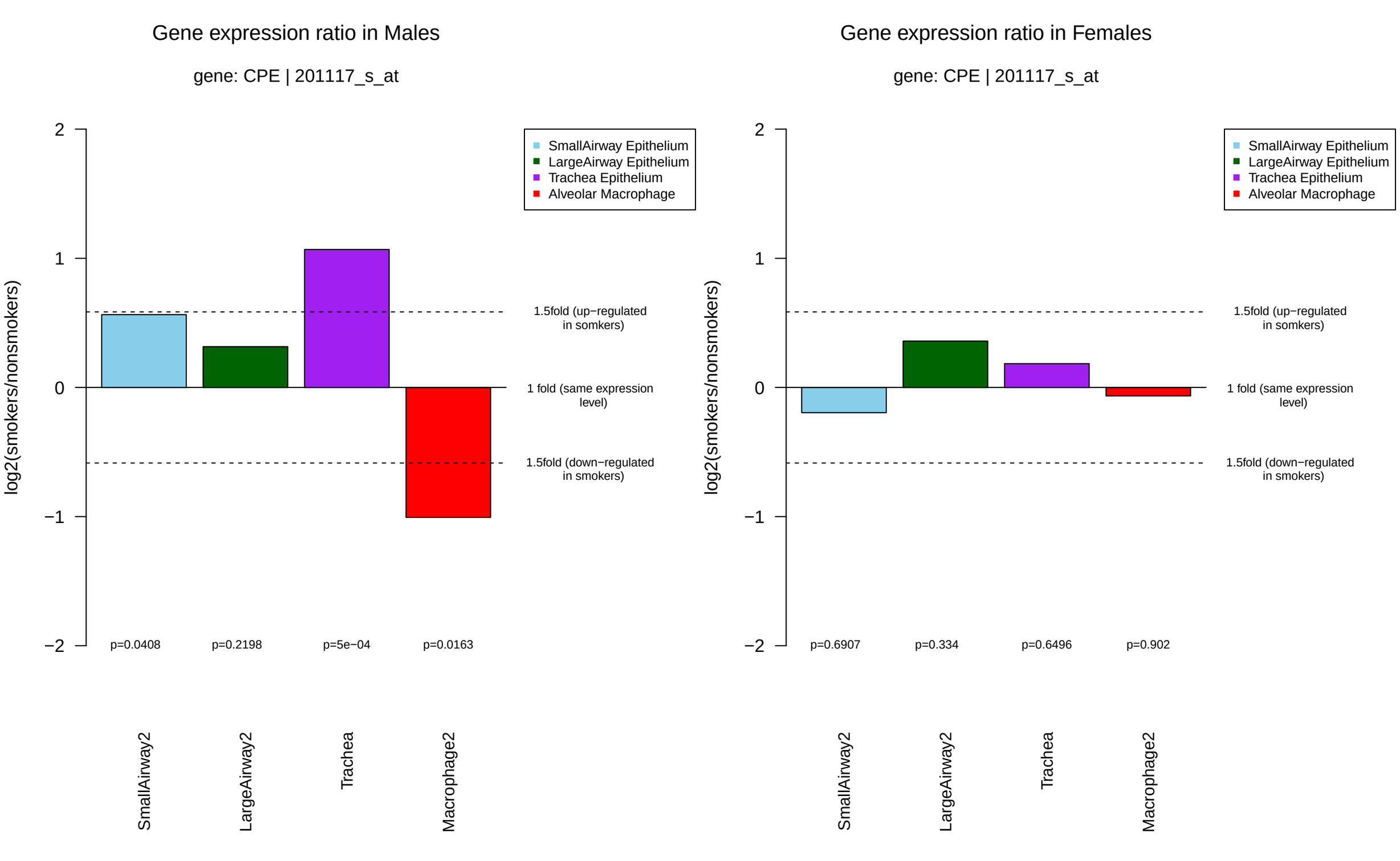
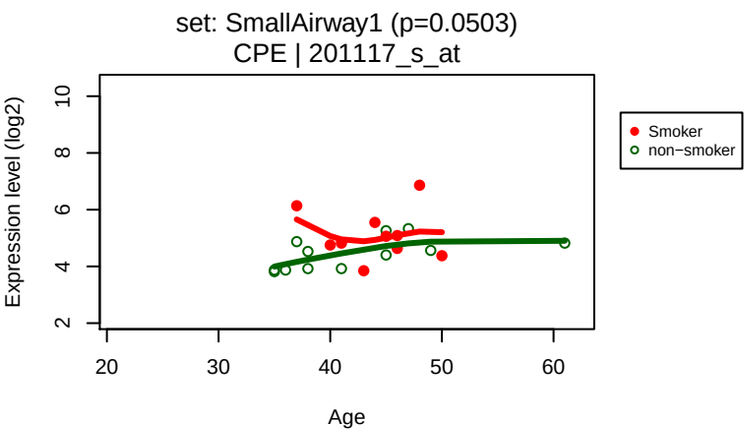
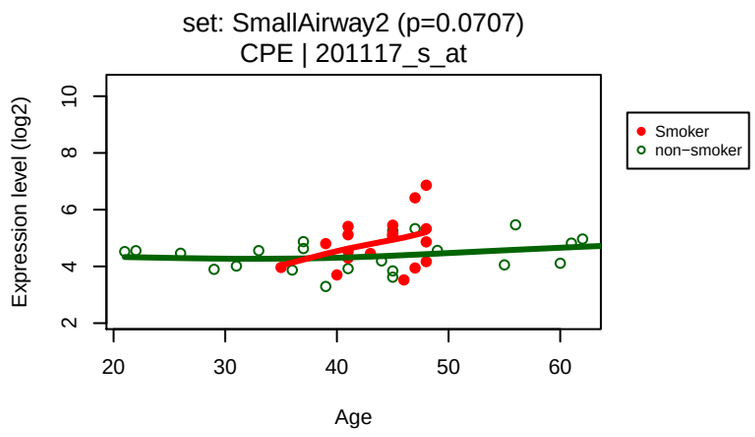
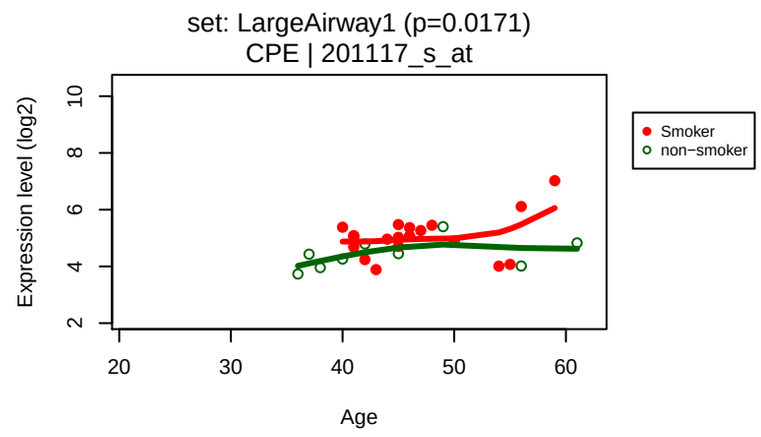

Gene expression of smokers and nonsmokers
gene: CPE | 201117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0503
p=0.0707
p=0.0039
p=0.0171
p=0.1146
p=5e−04
p=0.0385
p=0.0344
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

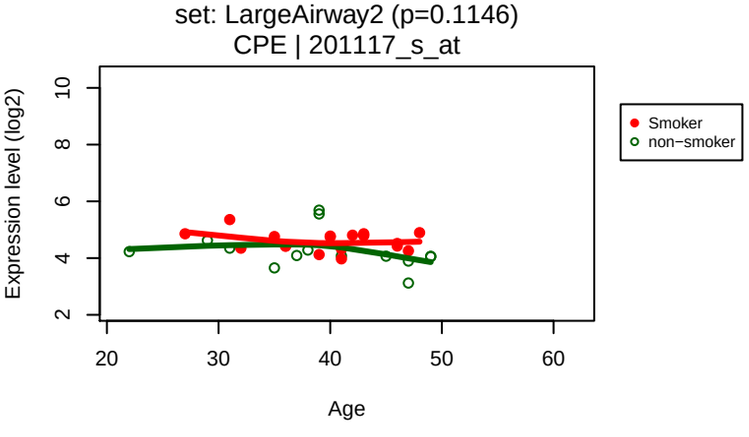

Gene expression ratio (smokers/nonsmokers)
gene: CPE | 201117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0503
p=0.0707
p=0.0039
p=0.0171
p=0.1146
p=5e−04
p=0.0385
p=0.0344
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

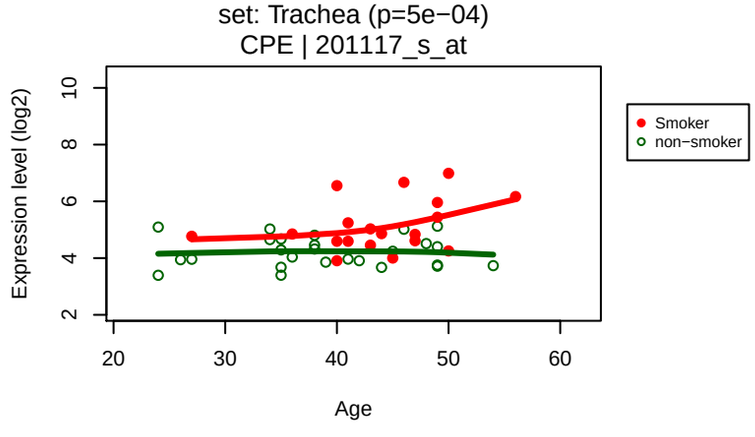

Gene expression ratio in Males
gene: CPE | 201117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0408
p=0.2198
p=5e−04
p=0.0163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CPE | 201117_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6907
p=0.334
p=0.6496
p=0.902
SmallAirway2
LargeAirway2
Trachea
Macrophage2

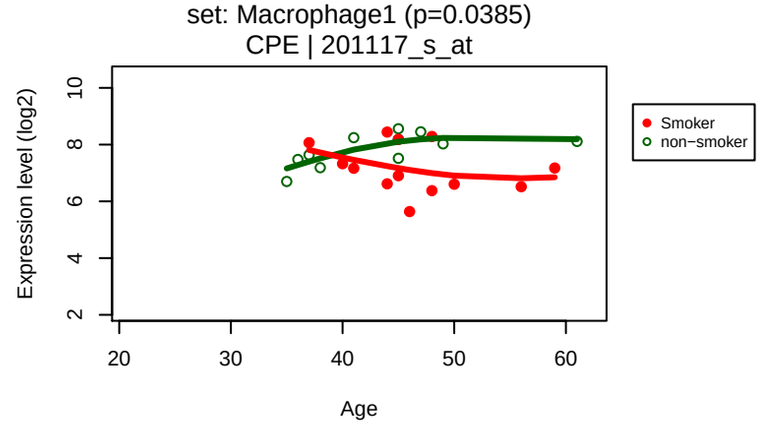

set: SmallAirway1 (p=0.0503)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker

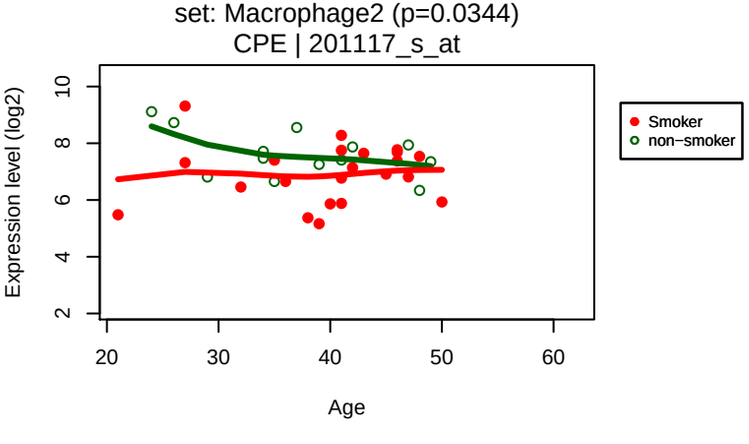

set: SmallAirway2 (p=0.0707)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker

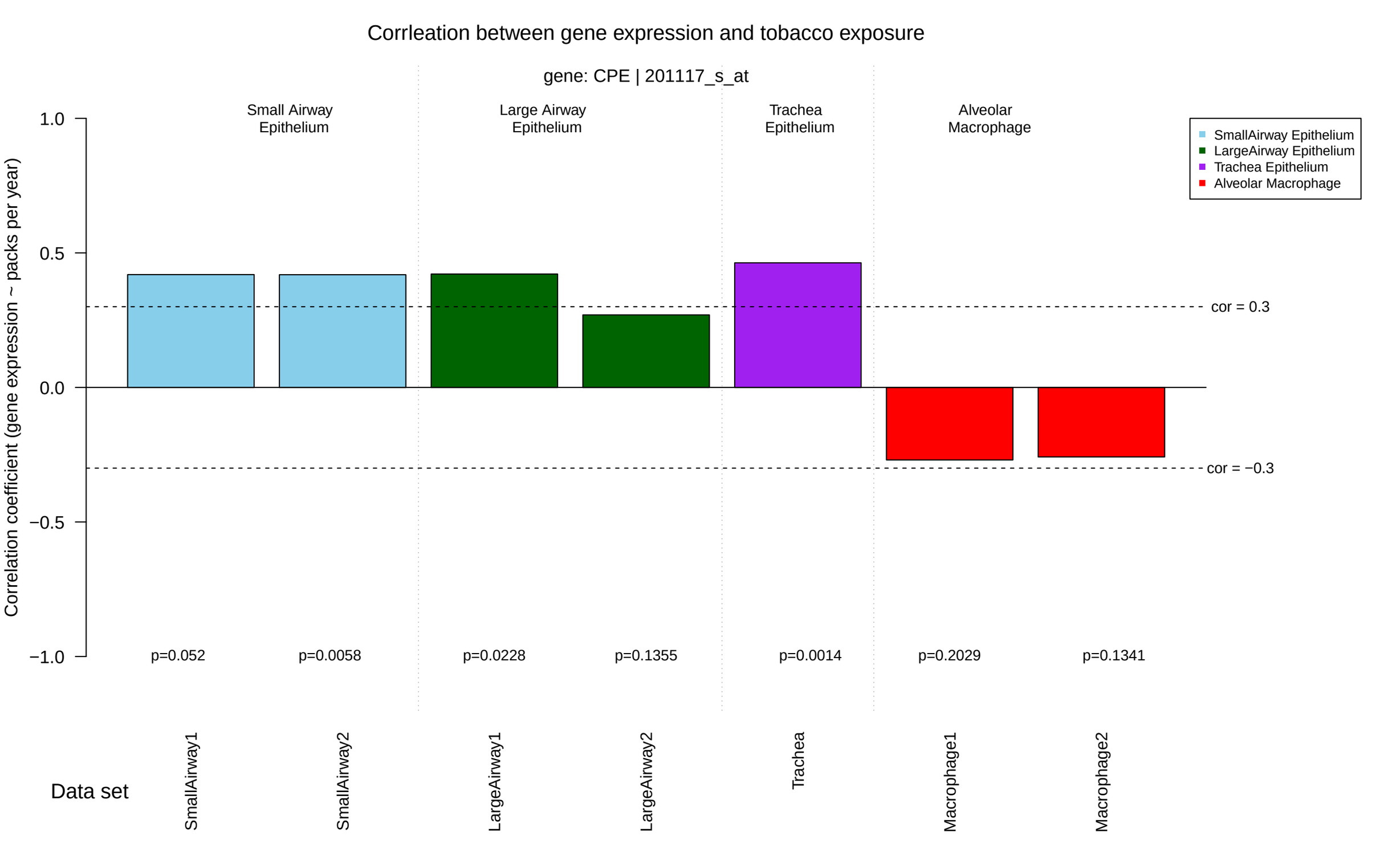

set: LargeAirway1 (p=0.0171)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1146)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0385)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0344)
CPE | 201117_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CPE | 201117_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.052
p=0.0058
p=0.0228
p=0.1355
p=0.0014
p=0.2029
p=0.1341
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2